[redacted]

additive inverses &
bitwise disjointness test

Is $\underline{CF=1}$: $(rax_0 \oplus rbx_0) = -(rbx_0 \oplus rcx_0)$

$\underline{CF=0}$: $(rax_0 \oplus rbx_0) \wedge (rbx_0 \oplus rcx_0) = 0$

in rsi $= (rax_0 \oplus rbx_0) + (rbx_0 \oplus rcx_0)$ overflows $\to 0$.

CF=0 always

unsigned overflow $\Updownarrow$ CF=1 $\Updownarrow$ rax=rbx

```
xor     rax, rbx
xor     rbx, rcx
mov     rsi, rax
add     rsi, rbx
cmovc   rax, rbx
xor     rax, rbx
cmp     rax, rsi
```

CF unchanged

rax, rbx $\to$ same sign $\Rightarrow$ MSB $\Downarrow$ 0

$rax = rax_0 \oplus rbx_0$

$rbx = rbx_0 \oplus rcx_0$

$rsi = rax$

$rsi = rsi + rbx$

$rsi = rax + rbx$

$= \underbrace{(rax_0 \oplus rbx_0) + (rbx_0 \oplus rcx_0)}_{\exists \text{ overflow ??}}$

CF $\Leftarrow$ stays the same ??

rbx $\leftarrow$ pivot register

$rax = rax \oplus rbx = rbx \oplus rbx = 0$

cmp (rax = 0, rsi ≥ 0) $\to$ 0

ELSE: $rsi = (rax_0 + rbx_0) + (rbx_0 + rcx_0)$ $\to$ no overflow $\Downarrow$ no cmove

$rax_0 = rax \oplus rbx = rbx_0 \oplus (rbx_0 \oplus rcx_0)$

$rax = \underbrace{rbx_0 \oplus rbx_0}_{} \oplus rcx_0$

$= 0 \oplus rcx_0$

$= rcx_0$

cmp $(rax = rcx_0,\ rsi = (rax_0 \oplus rbx_0) + (rbx_0 \oplus rcx_0))$

always clears CF

xor rax, rbx $\qquad rax = rax_0 \oplus rbx_0$

xor rbx, rcx $\qquad rbx = rbx_0 \oplus rcx_0$

CF untouched

mov rsi, rax $\qquad rsi = rax = rax_0 \oplus rbx_0$

add rsi, rbx $\qquad rsi = rsi + rbx$

$= (rax_0 \oplus rbx_0) + (rbx_0 \oplus rcx_0) = 0$

CF := 1 on this $\Updownarrow$

(CF := 1 $\Leftrightarrow$ unsigned(rsi) + unsigned(rbx) $\geq 2^{32}$)

$= \max(rsi)$

cmove rax, rbx $\qquad rax := rbx$

$:= rbx_0 \oplus rcx_0$ ELSE

$rax := rax_0 \oplus rbx_0$

$\Leftrightarrow CF = 1 \Leftrightarrow$ $\frac{rbx_0[2^{31}] = 0:}{rax_0[2^{31}] = 1 = rcx_0[2^{31}]}$ OR $\frac{rbx_0[2^{31}] = 1:}{rax_0[2^{31}] = 0 = rcx_0[2^{31}]}$

gives a (mod 64 bits vibe ...)

xor rax, rbx

cmp rax, rsi

$(rax_0 \oplus rbx_0) = -(rbx_0 \oplus rcx_0)$

$rax = (rbx_0 \oplus rcx_0) \oplus (rbx_0 \oplus rcx_0) = 0$ $\quad (a \oplus a = 0)$

CF = 1:

OR

CF = 0:

$rax = (rax_0 \oplus rbx_0) \oplus (rbx_0 \oplus rcx_0)$

ie additive inverse check

CF = 1:

cmp $(0, (rax_0 \oplus rbx_0) + (rbx_0 \oplus rcx_0))$ $\Rightarrow$ ZF = 1 ie same!

$\rightarrow$ if inverse CF, then it would only be 1 ×

CF = 0:

cmp $((rax_0 \oplus rbx_0) \oplus (rbx_0 \oplus rcx_0)$, $(rax_0 \oplus rbx_0) + (rbx_0 \oplus rcx_0))$

$+ \equiv \oplus \Leftrightarrow$ 2 operands share no common set bits

ie $A + B = A \oplus C$

$\Leftrightarrow$ B & C share no common set bits!

bitwise disjoint & orthogonally disjoint

adding & $\oplus$ (OVER $\oplus$'s) are equivalent ??

ie $B \oplus C$

$\|$

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ie {....}

ie $(rax_0 \oplus rbx_0) \wedge (rbx_0 \oplus rcx_0) = 0$

ex $0 \xrightarrow{CF=0} 0$